

Table 1. Sample and data architecture

Layer	Period / snapshot	Coverage	Purpose
Monthly ETF panel	Jan. 2010-Dec. 2025	4,459 ETF-months; 50 ETFs; 192 months	Pricing and cumulative adjustment
Fixed sustainable ETF universe	2025Q4 classification	72 included equity ETFs	Predefined historical reference set
Historical N-PORT match	2023Q4 / 2024Q4 / 2025Q4	56 / 63 / 66 exact series matches	Holdings reconstruction
Complete architecture panel	All three snapshots	56 ETFs with valid scores	Persistence and persistent averages
Boundary N-PORT validation	2020Q4 vs. 2025Q4	30 exact-series matches; 13 main-quality finds	Long-span temporal validation
Equity-corporate holdings	2023Q4 / 2024Q4 / 2025Q4	15,742 / 16,778 / 16,539 rows	Portfolio weighting
Firm fundamentals	FY2010-FY2025	27,312 firm-years; 2,174 firms	Lagged firm characteristics
Extreme CPU / joint stress	Monthly panel	20 / 17 months	Dynamic and threshold states

Note. The fixed ETF universe is determined from the 2025Q4 review and applied backward. The complete architecture count requires valid scores in every snapshot.